

Final Project Report

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Introduction

In this project we are trying to accomplish creating a unique, 3D printed truss bridge. While creating this bridge, we researched truss bridge designs, used CAD (Onshape) to model a bridge, and applied the method of joints to calculate the forces on each member. Truss bridges are especially useful compared to other types of bridges because they distribute weight and stress evenly throughout a structure and prevent deformation of members which are in tension or compression. By using a truss bridge design, we were able to see firsthand how truss bridges reduce material usage, have a high strength to weight ratio, and distribute forces well.

Comparison of Potential Design Influences

As mentioned before, truss bridges use interconnected triangle units to distribute loads to increase strength and reduce weight. There are many different types of truss bridges. For example, a Pratt truss uses diagonal members which slope inwards towards the center in tension, with vertical members in between in compression. The Howe truss (shown below in *Figure 1.2*) is the opposite, with sloped members pointing away from the center in compression, and verticals in tension. A Cable Stayed Bridge (shown below in *Figure 1.3*) is a combination design which uses cables to make an extremely stiff bridge. A Warren truss bridge (*Figure 1.1 and 1.4*), which we drew inspiration from, uses equilateral triangles with no vertical members to distribute loads.

Some “standout goals” for our design include using a pyramid shape at the top by creating members in 3 dimensions, having a high strength to weight ratio, to be easily assembled, and repeatable. By combining inspiration from all the images shown in *Figure 1*, we settled on a bridge design that uses non-equilateral triangles, with no vertical members, and in a pyramidal shape with a connecting bar on the top. With no vertical members, the load is distributed by the diagonals in compression and tension, and the bridge is lighter.



Figure 1. Truss bridge examples we drew inspiration from

1 (Top Left): Inglis Pyramid Bridge, England (Historic England, 2024).

2 (Top Right): Kuronoseto Bridge, Japan (Structurae, 2025).

3 (Bottom Left): Millau Viaduct, France (Structurae, 2025).

4 (Bottom Right): Brunsbüttel Viaduct, Germany (Structurae, 2025).

Justification of Design

The main goal of the design was to accomplish the goal of withstanding a load of 10 lbs. at the center with the least amount of material. From preexisting designs, the trusses were given the width of 0.3in (about 7.62mm) which was determined to give the optimal material strength ratio that we were looking for. The angles in each triangle were chosen to use the least number of triangles in the design to simplify the calculations and make it solvable. The length of the bridge was decided to be 14 inches since the bridge is going to span the gap of 10 inches, so we determined that we wanted at least 2 inches resting on the platform. The width was decided to be 4 inches, the narrowest option to limit the amount of material needed, and the height was also kept at 4 inches to conserve material.

To create uniqueness, fillets were used in certain locations to increase the stress distribution to increase its durability. From these design choices and since the bridge was cut in the middle, the center truss where the load will be applied, the thickness of that truss was increased to 0.5in so that the bridge could withstand more load since the weak point will be in the central beam where the load is going to be loaded.

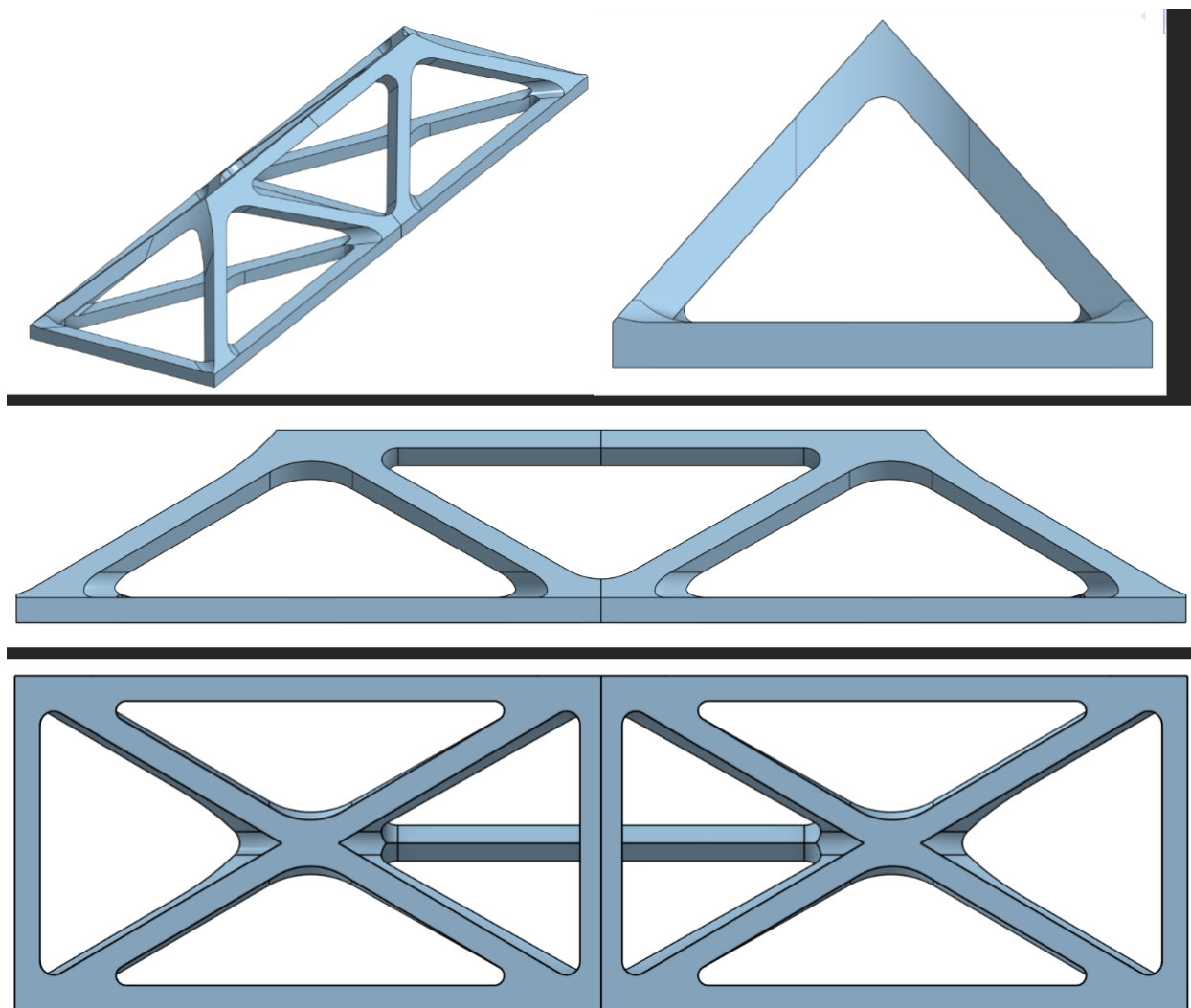


Figure 2. CAD Design (Onshape)

2a (Top Left): Isometric View

2b (Top Right): Front View

2c (Middle): Side View

2d (Bottom): Bottom View

Diagram of a truss structure with dimensions and forces:

- Dimensions: $l = 14 \text{ in}$, $w = 4 \text{ in}$, $h = 4 \text{ in}$.
- Text: "THRU GH & IJ symmetrical".
- Forces:
 - Upward forces of 2.5 lbs at nodes A, D, and F.
 - Downward forces of 5 lbs at nodes B and E.
- Nodes: A, B, C, D, E, F, G, H, I, J, K.
- Coordinate system: x , y , z .

To find the load of each individual member, we first need to find the reaction forces acting on the bridge. We know that the load of the bridge is -10 lbs. Since the corners of the bridge are going to rest atop the table, there is an upward (in the +z direction) reaction force at A, C, D, and F. Due to the bridge being symmetrical through BE and GH, the reaction force can be found by dividing 10 by 4 to yield 2.5 lbs.

$$A_z + C_z + D_z + F_z = 10$$

Figure 4. Solving using symmetry

For method of joints, at all the joints, there are more than 3 unknowns. Since there are only 3 equations, this means at this point, the system is statically indeterminate.

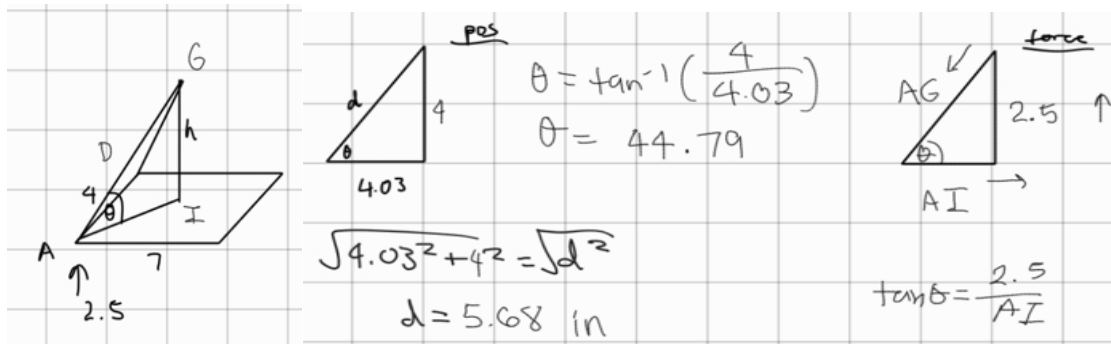


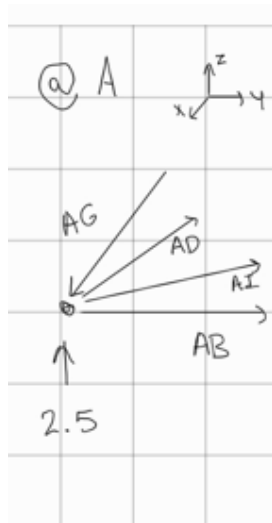
Figure 5. Sectioning system to solve AG and AI

However, if we separate the system into a pyramid like shown above in Figure 5 and further sectioning it off to a quarter at point A, leaving a right triangle AIG, the diagonal members AG and AI can be solved.

$$\begin{aligned}
 3.5^2 + 2^2 &= d_{AI}^2 \\
 d_{AI} &= 4.03 \\
 \Theta &= \tan^{-1}\left(\frac{4}{4.03}\right) \\
 \Theta &= 44.79
 \end{aligned}
 \qquad
 \begin{aligned}
 \sin \Theta &= \frac{2.5}{AG} \\
 AG &= \frac{2.5}{\sin \Theta} \\
 AG &= 3.55 \text{ lbs (compression)} \\
 AI &= \frac{2.5}{\tan \Theta} \\
 AI &= 2.52 \text{ lbs (tension)}
 \end{aligned}$$

Figure 6. Solving AG and AI

In Figure 6, using trigonometric relationships, we have found AG to be 3.55 lb in compression and AI to be 2.52 lb in tension.



$$r_{AI} = \langle -2, 3.5, 0 \rangle$$

$$\|r_{AI}\| = 4.03$$

$$\hat{AI} = \frac{r_{AI}}{\|r_{AI}\|} = \langle -0.5, 0.87, 0 \rangle$$

$$\vec{AI} = \hat{AI} * AI = \langle -1.25, 2.19, 0 \rangle \text{ (T)}$$

$$r_{AG} = \langle 2, -3.5, -4 \rangle$$

$$\|r_{AG}\| = 5.68$$

$$\hat{AG} = \frac{r_{AG}}{\|r_{AG}\|} = \langle -0.35, 0.62, 0.70 \rangle$$

$$\vec{AG} = \hat{AG} * AG = \langle -1.25, 2.19, -2.5 \rangle \text{ (C)}$$

$$\sum F_x = AG_x + AD + AI_x = 0$$

$$1.25 + AD - 1.25 = 0$$

$$AD = 0$$

$$\sum F_y = AG_y + AB + AI_y$$

$$-2.19 + AB + 2.19 = 0$$

$$AB = 0$$

$$\sum F_z = 2.5 - 2.5 = 0$$

Figure 7. Method of Joints: Solving for AD and AB and Joint A

Now that we have found members AG and AI, there are only 2 unknowns at joint A, making AB and AD solvable. First, we have to set up equilibrium force equations in the x and y

direction as seen in *Figure 7*. Using it, we calculate that AD and AB (and by extension BC, DE, EF and CF) are 0 force members.

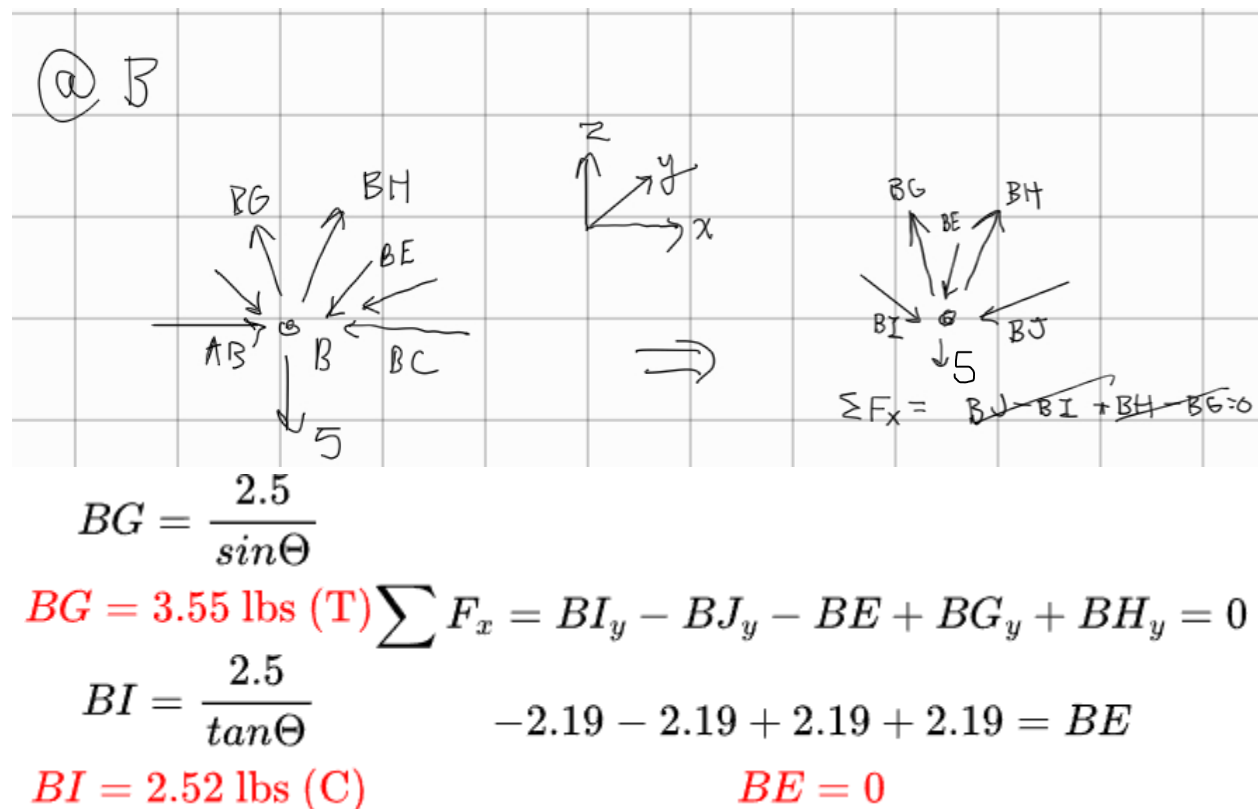
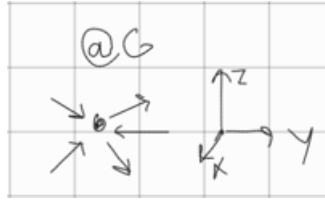


Figure 8. Solving for BG and BI, and using Method of Joints to solve for BE

To find BG and BI, we used a similar method in *Figures 6 and 7* but used point B as our reference point. As seen in *Figure 8*, since the 10 lb load was at location K, which is in the middle of member BE, we opted to treat the load as if it was split in half. This assumption made points B and E have a load of 5 lb downwards, due to symmetry. With this, we can calculate the members of BG and BI. Using the right triangle method used in *Figure 6*, we further break down the load of 5 lb in half to 2.5 lb due to there being 2 diagonal members in x, y, and z directions at point B instead of one found at point A. We found that BG = 3.55 lb in tension and BI = 2.52 lb in compression. To note, the members have the same magnitude as their counterparts in *Figure 6* but have the opposite load stress type. Finally, using the equilibrium equation in the y-direction (because the x and z directions cancel out to 0), we found that BE is also a 0-force member.



$$\sum F_y = -GH + AG_y + BG_y + DG_y + EG_y = 0$$

$$2.19 + 2.19 + 2.19 + 2.19 = GH$$

$$GH = 8.76 \text{ lbs (C)}$$

Figure 9. GH calculation

Finally, to calculate the last member of GH, we use method of joints at Joint G. Using the equilibrium equation in the y-direction (because the x and z directions cancel out to 0), we found that GH is 8.76 lb in compression.

To summarize, we found that:

$$AG = DG = CH = FH = 3.55 \text{ lb (compression)}$$

$$BG = EG = BH = EH = 3.55 \text{ lb (tension)}$$

$$AI = DI = CH = FH = 2.52 \text{ lb (tension)}$$

$$BI = EI = BH = EH = 2.52 \text{ lb (compression)}$$

$$GH = 8.76 \text{ lb (compression)}$$

$$AB = BC = DE = EF = 0$$

$$AD = CF = 0$$

$$BE = 0$$

One disclaimer we would like to add is that the right triangle trigonometric method we used in *Figures 5, 6, and 7* or the splitting 10 lb in half at points B and E in *Figure 8* are assumptions on how to simplify the calculations using the coursework we worked on in this course. One point of concern is that we are unsure that in *Figures 5, 6, and 7*, the diagonal members lying in the xy plane (AI, BI, etc.) are able to be calculated using the right triangle trigonometric method because it would always yield a value that would be opposite of the diagonal going in the xyz direction (AG, BG, etc.), therefore making the floor members that only go in one direction be zero-force members. However, if we did not employ this method, we found that the floor members are statically indeterminate, and we take member BE to be -2.5 lb. Due to us deciding that we should at least try to obtain a load value for all the members, we

assume BE = 0 and employed the right triangle trigonometric method, concluding the rest of the floor members are 0-force members since they form a 2D truss.

Since stress = load/adjusted area ($\sigma = \frac{P}{A}$), the member GH attaching the two truss pyramids has the largest load of 8.76 lb. We know this member has the largest load and therefore the most stress since the cross-sectional area of the members is uniform through the truss bridge. The stress in this case is 486.7 psi.

$$\text{Adjusted Area} = 0.3 \text{ in} \times 0.3 \text{ in} \times 0.2 = 0.018 \text{ in}^2$$

$$\sigma = 486.7 \text{ psi} = 3.355 \text{ MPa}$$

GH would break first in an ideal situation, but member BE is attached together using tape. While super glue was preferred, the surface area was too small to connect it and warped due to printing errors. Essentially glue was not an option, so we used tape to combat this. Member BE has ideally 0 load, but under stress it is likely to break first.

Conclusion

Overall, our bridge design was a success. The calculations were a huge help in finalizing the design. We realized that in our initial design, there were several zero-force members that were unnecessary and added unneeded weight and material to our bridge.

After seeing the calculations, the bridge was very stable, carrying evenly distributed weight, except for the top beam. We found the top horizontal bar to be 8.76 lb in tension. This horizontal bar is important not only because it carries a substantial load, but it improves stability and stiffness, preventing the bridge from buckling as more weight is added to the bridge during load testing. Additionally, fillets were used here to increase the stress distribution to increase its durability.

If the bridge were loaded until failure, it would approximately fail at 35 lb is our estimate considering the infill, tensile strength of the material, the weakest member with the most stress. With a 20% infill and only a 0.09 in cross-sectional area, our team was very focused on creating a high load/weight ratio for an optimal design. With this constraint, our bridge holds less weight. PETG's ultimate tensile strength (UTS) in this case was 17.43 MPa (2528 psi), based on a 20% rectilinear infill (Kumaresan, 2023). Taking all these factors into account, we were able to calculate an estimated load of 45 lb. But accounting for bending, the tape, and printing errors we estimated to failure load to be 35 lb.

$$\frac{P_{fail}}{0.018 \text{ in}^2} = 2528 \text{ psi} = 45 \text{ lb} \rightarrow 35 \text{ lb}$$

The design can be improved by distributing the weight in the x-component better. In the current design, the member GH bears the majority of the load, making it a critical weak point due to high load and resulting stress. This single member carrying so much force makes it susceptible to failure, buckling, fracturing, deformation, etc. To improve the current design, adding more members to help redistribute the load would go a long way. For example, adding members at point G and point H and connection to the ends of the bridge would help distribute the load.

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